

Longfei Li

Institutes of Science and Development, Chinese Academy of Sciences

 GitHub |  Google Scholar |  lilongfei22@mailsucas.ac.cn |  +86 156 2869 5648
 Personal Website |  Beijing, China

Education

2022.09 – now **University of Chinese Academy of Sciences (UCAS)** Beijing, China
Successive postgraduate and doctoral program in Management Science and Engineering
Supervisors: [Xiaolei Sun \(Prof. Sun\)](#), [Qiang Ji \(Prof. Ji\)](#), GPA: 3.79/4.0
Research Areas: Risk management, Resilience assessment, Big data decision-making
2018.09 – 2022.06 **Shandong University (SDU)** Weihai, China
B.B.A. in Human Resource Management
Mentor: [Chao Liu \(Prof. Liu\)](#), GPA: 3.9/4.0, Rank: 2/78
Recommended for Master's Admission without Entrance Exam, Outstanding graduates

Publications

Category	Quantity
Published Journal Articles (peer-reviewed)	11
Book Chapters	3
Policy Articles	5
Software Copyright	1
Total	20

Undergoing Review: 3 | Working Papers: 3

Accepted & Published

- [1] Xiaolei Sun, [Longfei Li](#), Yiran Shen, Yingjie Sheng, Dayong Zhang*, Qiang Ji*. (2025). Climate risk expressions of public views and government responses in China. *Scientific Data*, 12(1), 877. (SCI, JCR Q1, IF=6.9)
- [2] [Longfei Li](#), Junnan Zhang, Chao Liu*, Yuyue Guan, Wan Kang. (2025). Knowledge creation network in response to crisis: A static-dynamic framework for innovation network resilience from dual-risk perspective. *Journal of Innovation & Knowledge*, 10(6): 100825. (SCI, JCR Q1, IF=15.5)
- [3] [Longfei Li](#), Xiaolei Sun*, Ting Chen, Yiming Zhao. (2026). Structural analysis and knowledge graph construction of policy elements with “tool-relationship-object” framework. *Chinese Journal of Management* (管理学报), Accepted. (CSSCI, in Chinese)
- [4] Xiaolei Sun, [Longfei Li](#), Lili Yuan, Yu Song, Dayong Zhang*. (2026). Promoting the joint development of digital standards for the Belt and Road initiative: achievements, challenges, and future strategies. *Bulletin of the Chinese Academy of Sciences* (中国科学院院刊), 41(2), 381–392. (CSSCI, in Chinese)
- [5] Chao Liu, [Longfei Li](#)*, Keyi Lu. (2025). Resilience characteristics and evolution dynamics of China's integrated circuit industry innovation network. *Technology Analysis & Strategic Management*. Online: <https://doi.org/10.1080/09537325.2025.2539474>. (SSCI, JCR Q2, IF=2.8)
- [6] [Longfei Li](#), Xiaolei Sun, Weilan Suo*. (2023). Construction of risk response scenarios for the emergency material support system. *Procedia Computer Science*, 221, 979–983. (EI)
- [7] Yingjie Sheng, Xiaolei Sun, Hubin Ma, [Longfei Li](#)*. (2025). Multidimensional deconstruction of policy tools: A quantitative assessment framework based on frequency effectiveness and positional. *Procedia Computer Science*, 266, 222–228. (EI)
- [8] Xiaolei Sun, Yiran Shen, [Longfei Li](#), Weilan Suo*. (2023). Returns, volatility and investor sentiment networks: Evidence from China's listed energy companies. *Journal of Systems Science and Mathematical Sciences* (系统科学与数学), 43(7), 1713–1729. (CSCD, in Chinese)
- [9] Weilan Suo, Wenjie Xu, [Longfei Li](#), Xiaolei Sun*. (2025). A novel data-intelligence-driven three-stage dynamic model for resilience assessment in an emergency material support system. *International Journal of Critical Infrastructure Protection*, 51, 100804. (SCI, JCR Q1, IF=5.3)
- [10] Yuyue Guan, [Longfei Li](#), Chao Liu*. (2023). Resilience characteristics and driving mechanism of urban collaborative innovation network—A case study of China's new energy vehicle industry. *Systems*, 11(5), 214. (SSCI, JCR Q1, IF=3.1)

- [11] Xinyang Guo, Longfei Li, Zongshui Wang, Hong Zhao*. (2026). Pattern recognition of innovation partnerships in China integrated circuit industry: Application of network motif analytics. *Systems*, 14(4): 433. (SSCI, JCR Q1, IF=3.1)

Undergoing Review

- [12] Longfei Li, Xiaolei Sun*, Weilan Suo, Qiang Ji. Modeling cascading risk and response mechanisms under extreme climate disasters: A symbolic event logic graph framework. Undergoing review in *International Journal of Disaster Risk Science*.
- [13] Xiaolei Sun, Longfei Li, Jing Zhang, Weilan Suo*. Quantitative evaluation of the effectiveness of the collaborative response system for major emergencies based on risk transmission. Undergoing review in *Operations Research and Management Science* (运筹与管理, in Chinese).
- [14] Longfei Li, Xiaolei Sun*. Task reassignment and robustness optimization of emergency response network for cascading risks under extreme climate disasters. Undergoing review in *Reliability Engineering & System Safety*.

Working Paper

- [15] Deep learning classification of disaster information based on multimodal data.
- [16] Policy synergy in the governance of CO₂ and non-CO₂.
- [17] Climate response action model validation and reflection learning agent.

Policy Articles

- [18] Longfei Li, Xiaolei Sun, Yi Ding, Lili Yuan. (2023). FEMA national risk index for natural disasters. *China Emergency Management* (中国应急管理), 09, 50–53. (in Chinese)
- [19] Longfei Li, Yuxia Lü, Lili Yuan. (2024). Pathways and strategies for advancing the modernization of emergency medical and health material support systems. *Disaster Reduction in China* (中国减灾), 21, 30–33. (in Chinese)
- [20] Wenjie Xu, Longfei Li, Weilan Suo, Xiaolei Sun. (2024). Optimization pathways for emergency material support systems in response to compound extreme events. *Disaster Reduction in China* (中国减灾), 21, 30–33. (in Chinese)
- [21] Xiaolei Sun, Jing Zhang, Longfei Li. (2024). Major emergency response for complex scenarios. *Disaster Reduction in China* (中国减灾), 11, 42–44. (in Chinese)
- [22] Longfei Li, Xiaolei Sun. (2024). Smart Life Line: A new paradigm for emergency logistics distribution driven by artificial intelligence. *China Grain Economy* (中国粮食经济), 11, 59–61. (in Chinese)

Book Chapters

- [23] Yiran Shen, Longfei Li. (2025). Risk-return analysis of China's traditional energy and new energy industries. In Xiaolei Sun & Qiang Ji (Eds.), *Energy and Critical Mineral Security in China* (pp. 151–178). Springer. https://doi.org/10.1007/978-981-96-7147-2_7.
- [24] Longfei Li, Xiaolei Sun, Weilan Suo. (2025). Construction and structured expression of risk response scenarios for major emergencies. In Xiaolei Sun, Weilan Suo, & Jing Zhang (Eds.), *Resilience Research on Emergency Material Assurance Systems from the Perspective of Risk Governance* (pp. 17–32). Science Press. (in Chinese)
- [25] Xiaolei Sun, Hubin Ma, Longfei Li, Wenjie Xu. (2025). Four directions for AI-enabled emergency material reserve management. In Xiaolei Sun, Weilan Suo, & Jing Zhang (Eds.), *Resilience Research on Emergency Material Assurance Systems from the Perspective of Risk Governance* (pp. 218–246). Science Press. (in Chinese)

Software Copyright

- [26] Qiang Ji, Kun Guo, Pengxiang Zhai, Longfei Li, Liyuan Lan, Yu Liu, Yuyue Yang. LNG Import Price Multi-scale Decomposition and Dynamic Evolution Pattern Recognition System V1.0 [CP]. Registration Number: Soft Copyright Registration No. 16705320, 2025. (in Chinese)

Research Methods with Selected Publications

Note: Clickable numbers link to the corresponding publication in the list above.

Machine Learning	Statistical machine learning; LLMs; Multimodal. [[1], [3], [7], [12], [13], [14], [15], [17], [23], [24], [25], [26]]
Social Network	Network Dynamics; Network Modeling & Inference; Structure Mining. [[2], [3], [5], [7], [8], [10], [11], [12], [13], [14], [16], [23], [24], [25]]
Multiple Attribute Decision Making	Criteria integration and ranking. [[7], [9], [14]]
Qualitative Analysis	Positivism, Interpretivism, QCA, Opinion articles. [[4], [6], [12], [18], [19], [20], [21], [22], [24], [25]]
Microeconometrics and Causal Inference	Regression, Robustness, Endogeneity, Event analysis, Survival analysis. [[8]]
Optimization Theory & Methods	Simplex method, sensitivity test, duality theory, convex optimization, RO, CCP, two-stage stochastic programming model.

Talks and Presentations

June 14, 2026	Policy synergy in the governance of CO ₂ and non-CO ₂ . Conference talk, Xidian University, Xi'an, China. <i>2026 International Conference on Climate and Energy Finance (ICEF 2026)</i> .
June 07, 2026	Unraveling Cascading Risk and Response Mechanisms in Extreme Rainstorms: A Computable Framework for Mapping Risk-Response Relationship. Conference talk, Beihang University, Beijing, China. <i>The 10th Annual Conference of the Emergency Management Systems Engineering Professional Committee, Chinese Society of Systems Science</i> .
May 31, 2026	A Task Reassignment-Oriented Framework for Analyzing Response Resilience to Extreme Climate Disasters under Cascading Risks. Conference talk, Sichuan University, Chengdu, China. <i>The 11th Interdisciplinary Forum on Risk, Disaster, and Crisis</i> .
August 17, 2025	Multidimensional deconstruction of policy tools: A quantitative assessment framework based on frequency effectiveness and positional. Conference talk, Online talk, Rutgers Business School, New Brunswick, Piscataway, USA. <i>Presented in 12th International Conference on Information Technology and Quantitative Management (ITQM 2025)</i> .
August 02, 2025	Policy Element Extraction and Structural Analysis Based on Large Language Models. Best Paper . Conference talk, Shandong University, Jinan, China. <i>The 5th Annual Conference of the Data Science and Knowledge Systems Engineering Professional Committee, Chinese Society of Systems Science</i> .
June 02, 2024	A public complain based climate social susceptibility and government response index for China. Conference talk, Shandong University, Weihai, China. <i>Presented in 2024 International Conference on Climate and Energy Finance (ICEF 2024)</i> .
August 13, 2023	Construction of risk response scenarios for the emergency material support system. Conference talk, Online talk, Oxford, UK. <i>Presented in 10th International Conference on Information Technology and Quantitative Management (ITQM 2023)</i> .

Research Projects

2025 – 2029, member	Ecological Security Smart Decision-making , JYB2025XDXM904, Fundamental and Interdisciplinary Disciplines Breakthrough Plan of Ministry of Education of the People's Republic of China
2026, member	Research on the Knowledge Integration and Implementation Mechanism for the Transformation of Qinghai-Tibet Scientific Expedition Results into Regional Development , Science and Technology Department of Qinghai Province
2026, member	Research on Layout Optimization of Central Emergency Rescue and Relief Materials Reserve Warehouse and Emergency Guarantee Mechanism for Responding to “Three Disconnections” Situations , National Food and Strategic Reserves Administration
2025 – 2026, member	Research on the Central-Local Collaborative Allocation Mechanism of Emergency Materials under Multiple Risk Scenarios , China Grain Research and Training Center
2024 – 2026, member	Game countermeasure of science and technology strategy supported by digital technology , 2022ZD0116407, Science and Technology Innovation 2030 – Major Project
2024 – 2026, member	Construction of a Forecast Model Database for China's Spot Prices of Imported LNG , CNOOC Gas and Power Group Co., Ltd.
2023 – 2025, member	Intelligent emergency material logistics deployment technology and application for major emergencies , 2021ZD0114201, Science and Technology Innovation 2030 – Major Project
2024, member	Research on the Central Emergency Rescue and Disaster Relief Material Allocation Guarantee System and Risk Response Capacity , National Food and Strategic Reserves Administration
2023, member	Research on the Scenarios and Operating Strategies of the Market Regulation Mechanism for Key Minerals , China Grain Research and Training Center
2021 – 2022, member	Research on optimization of innovation factor structure of old industrial base on supply-side structural reform , 21BGL304, National Social Science Foundation of China

Awards

2025	Cheng Siwei Outstanding Student Award, UCAS Education Foundation
2023, 2024, 2025	Merit Student, UCAS
2023	First Prize, 3rd “Green Apple Cup” Management Graduate Thesis Competition, Rank 1/78, CAS , Institutes of Science and Development
2023	Outstanding Prize (the highest award), 9th China National College Students Competition on Energy Economics, CSCOPEM
2022	Outstanding Graduates, SDU
2020, 2021	Social Practice Excellent Team in Shandong Province (2020 Social Practice Excellent Individual in Shandong), Department of Education in Shandong Province
2021	Bronze Prize, 12th “Challenge Cup” National College Students' Business Plan Competition (CCYL , CAST , CASS , MOE , ACSF)
2021	Excellent Rating, Research Assistant Evaluation, Rank 1/141, SDU , School of Business
2020	Second Prize, 16th “Challenge Cup” National College Students' Academic Science and Technology Competition (CCYL , CAST , CASS , MOE , ACSF)

English Proficiency & Overseas Study Experience

English Proficiency	CET-4; CET-6; IELTS (6.5)
Overseas Study	Technical University of Munich (Visiting and study program, 2019.08, Digital Transformation and Industry 4.0) National University of Singapore (Online, 2021.12, Data Analytics and Mathematical Statistics, Distinction)

Review Service for Journals

Humanities & Social Sciences Communications, Journal of Innovation & Knowledge, Economic Analysis and Policy, Technology Analysis & Strategic Management, Habitat International, Applied Spatial Analysis and Policy, Systems Research and Behavioral Science, Science and Public Policy, Journal of Industrial Ecology, Growth and Change.